Interactions

Interaction patterns find shapes on different layers whose regions overlap , candidates for via/contact checks

The idea

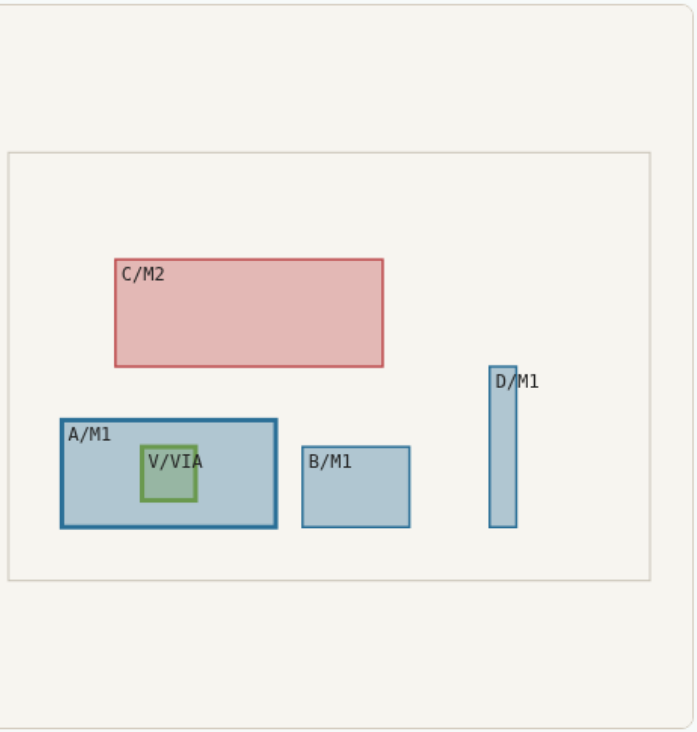
- V overlaps A (VIA∩M1)
- GOLDEN: at least the (A,V) pair

Cross

MULTI-LAYER INTERACTIONS

Cross

Step 1 / 5 · cross · module03-03-multi-layer-interaction



Explanation

Only different-layer pairs qualify.

- cross

```
≠layer
```

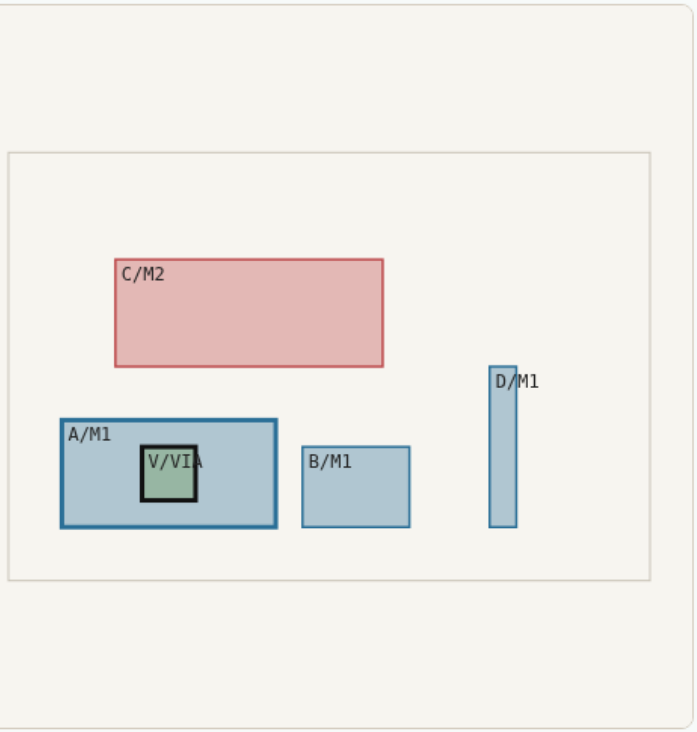
Cross

Overlap

MULTI-LAYER INTERACTIONS

Overlap

Step 2 / 5 · ov · module03-03-multi-layer-interaction



Explanation

Axis-aligned rectangle overlap test.

- AABB

```
overlap
```

Overlap

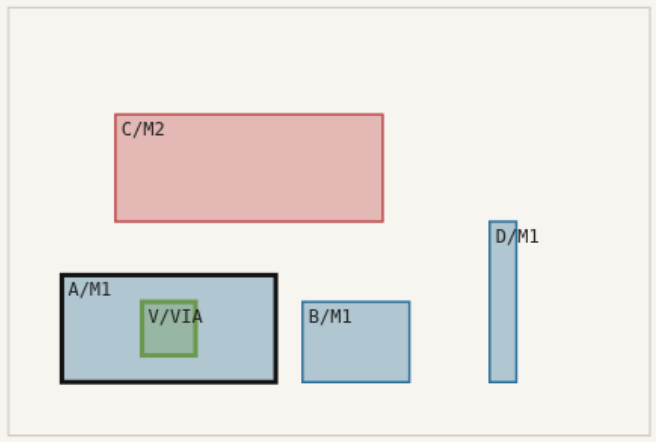
AV

MULTI-LAYER INTERACTIONS

AV

Step 3 / 5 · av · module03-03-multi-layer-interaction

A-V



Explanation

Pair A–V is present.

- A–V

yes

AV

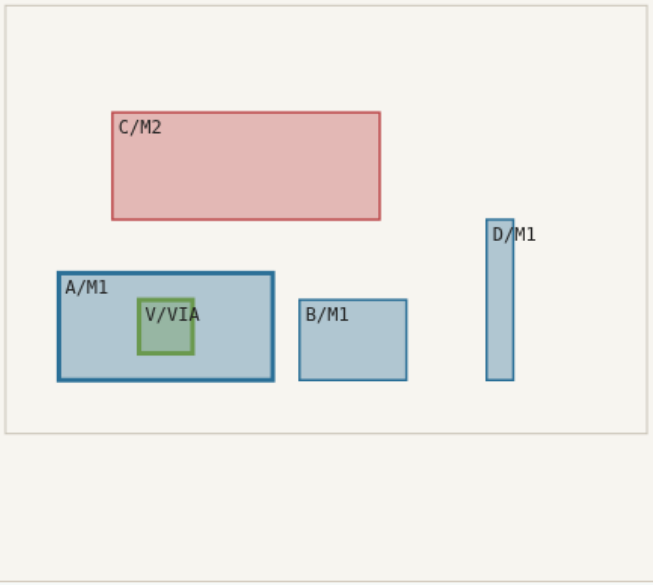
List

MULTI-LAYER INTERACTIONS

List

Step 4 / 5 · list · module03-03-multi-layer-interaction

pairs=1



Explanation

multi_layer_interactions returns pairs.

- list

```
n=1
```

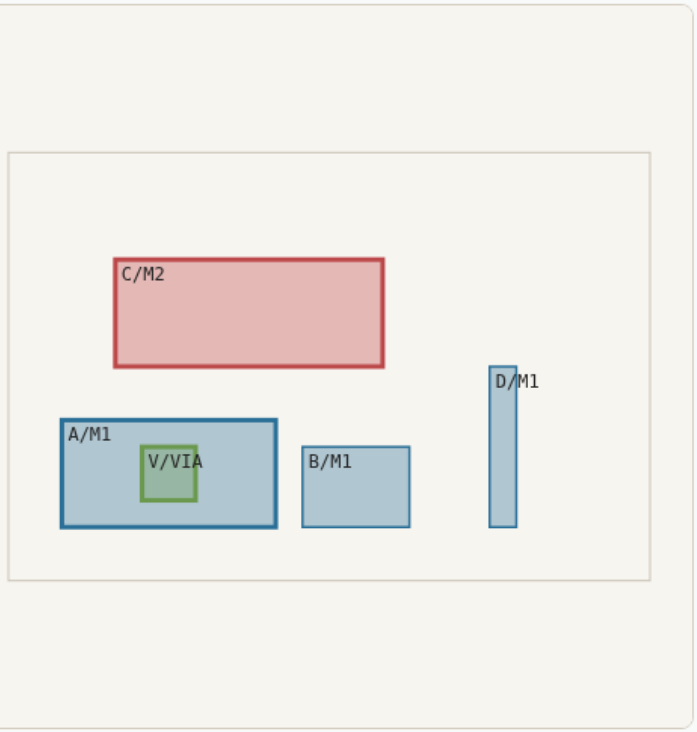
List

Use

MULTI-LAYER INTERACTIONS

Use

Step 5 / 5 · use · module03-03-multi-layer-interaction



Explanation

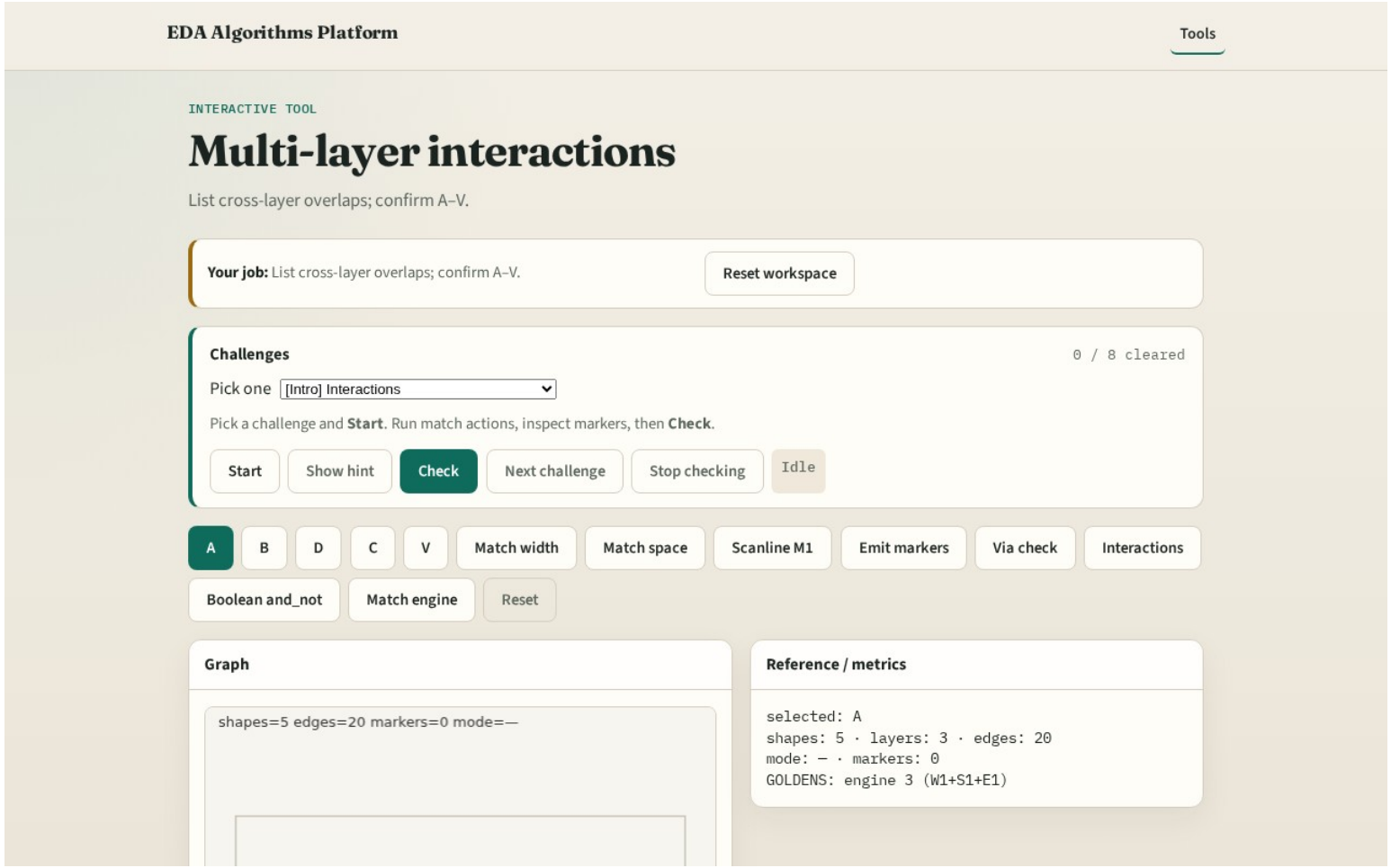
Interactions feed enclosure and boolean patterns.

- next

```
boolean
```

Use

Browser lab track



Browser lab starter

Implement track

- Call multi_layer_interactions(); assert ('A','V') in pairs

Pitfalls

- Including same-layer overlaps
- Using center distance only

Your turn

- Next: layered boolean